📦 CoCo CLI Hack · GCC Edition

Supply Chain Governed Conversational Analytics

A governed ontology on Snowflake that lets anyone ask supply-chain questions in plain English — and get **one trusted number**, with the definition attached.

Snowflake | Cortex Analyst | Semantic Views | Dynamic Tables | Streamlit | Built with CoCo

Challenge: Supply Chain Ontology & Governed Conversational Analytics

THE PROBLEM

Everyone asks the same question and gets a different number

In supply chain, "What's our on-time delivery rate?" has four different answers depending on who you ask.

🧩 Fragmented definitions

Procurement, Logistics, Finance and Sales each compute OTD, fill rate and landed cost differently — no shared truth.

⏱️ Slow to answer

Every question becomes an analyst ticket and a bespoke SQL query. Insight lags the decision.

🔍 No lineage or trust

Numbers arrive with no definition, no data-quality guarantee, and no way to audit how they were derived.

🗣️ Not self-serve

Business users can't explore data themselves — they wait on a technical team for every slice.

The cost: inconsistent decisions, wasted analyst hours, and disputes over whose number is right.

THE SOLUTION

A governed ontology you can talk to

One canonical model, four sanctioned metrics, and a natural-language interface grounded in it.

1 · Governed ontology

A canonical Supply Chain model (Supplier → Part → Plant → Shipment → Order → Customer) built as a Bronze→Silver→Gold medallion on Snowflake Dynamic Tables.

2 · One definition per metric

OTD, Fill Rate, Days of Inventory, Landed Cost — each with a single sanctioned definition, stored as data and enforced in the semantic view.

3 · Ask in plain English

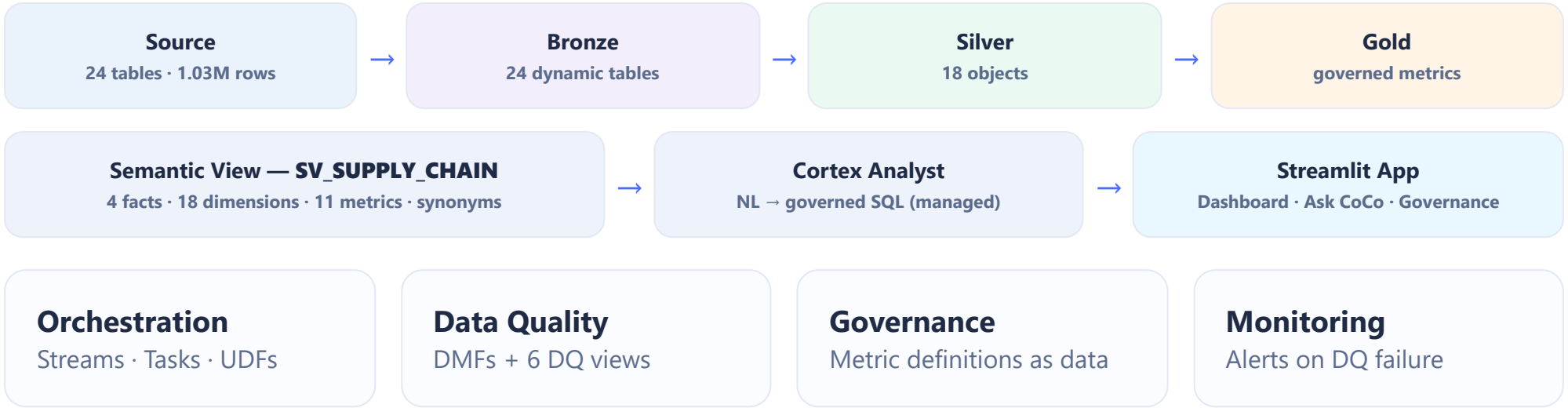
Cortex Analyst turns natural-language questions into governed SQL over the semantic view, so every persona gets the same trusted answer.

Outcome: a Streamlit app where a business user types a question and gets an answer, the SQL behind it, a chart, and the governed definition — no analyst ticket required.

ARCHITECTURE

Source → Medallion → Semantic View → Cortex Analyst → App

All on Snowflake. Governance, data quality and orchestration run across every layer.



Deterministic synthetic data (hash-seeded) — a judge can reproduce every number exactly.

GOVERNANCE

Four metrics, one sanctioned definition each

Definitions are stored as queryable data *and* enforced in the semantic view — this is what makes every answer citable.

METRIC	GOVERNED DEFINITION (SUMMARY)	GRAIN
On-Time Delivery	Lines where receipt date ≤ promised date. Excludes cancelled & not-yet-due.	Month × Supplier × Region × Category
Fill Rate	Units filled on first attempt ÷ units ordered. No backorder credit.	Month × Service Tier × Category × WH
Days of Inventory	On-hand ÷ average daily demand (part-warehouse grain).	Snapshot × Part × Warehouse
Landed Cost / unit	Purchase + freight + insurance + duties + handling, per unit (qty-weighted).	Month × Supplier × Route × Category

✓ **Cross-persona consistency** — ask OTD five different ways, get one number, every time, with the definition attached. That is the difference between this and a dashboard.

THE PROTOTYPE

One app, three tabs, complete story

Streamlit-in-Snowflake — see the numbers, ask anything, trust the result.

📊 Dashboard

4 KPI tiles + monthly on-time-delivery trend. Loads instantly (cached). The demand-shock dip is visible at a glance.

💬 Ask CoCo

Natural-language chat. Type any supply-chain question → governed SQL → answer + the generated SQL + a chart.

✅ Governance

Live data-quality scorecard (42/42 checks pass) and the governed metric-definition catalog.

78.1%	90.3%	7.0	$113
On-Time Delivery	Fill Rate	Days of Inventory	Landed Cost / unit

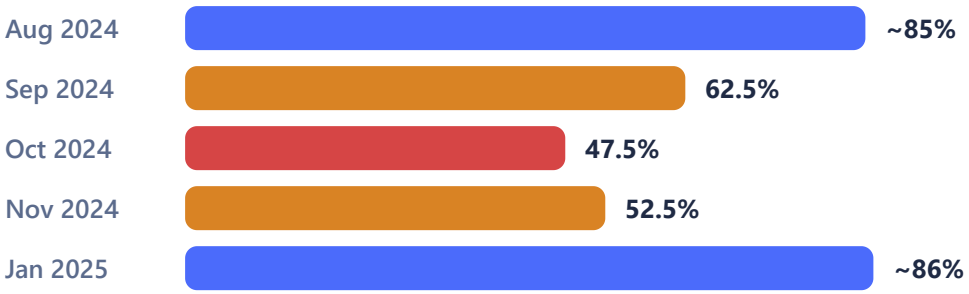
Runs on the warehouse runtime; Cortex Analyst called via the in-Snowflake REST bridge over **SV_SUPPLY_CHAIN**.

DEMO STORY

The data tells a real supply-chain story

A demand-shock scenario is embedded in the data — the app surfaces it in one question.

"Show on-time delivery by month"



Recovery by Jan 2025 after the shock.

What the app reveals

- A supplier-disruption + demand-spike scenario drives OTD from ~85% to a **47.5% trough**.
- The line chart makes the dip obvious — no analyst required.
- Follow-ups ("which region is worst?", "which supplier?") drill straight in.
- Every answer is the **same governed number** regardless of who asks.

IN ACTION

Ask anything — get a governed answer

Verified live against the semantic view. Same question, same number, every persona.

NATURAL-LANGUAGE QUESTION	ANSWER
What was the overall on-time delivery rate?	**74.26%**
Fill rate by customer service tier	Gold 90.3 · Platinum 89.7 · Silver 90.9 · Standard 90.3
Days of inventory by part category	6.1 – 8.3 days
Average landed cost per unit by supplier	$390 – $548 (top suppliers)
Which region has the worst OTD?	**Africa — 73.97%**
Show on-time delivery by month	Trend chart with the Sep–Nov 2024 dip

Each answer ships with the generated SQL and a chart — transparent and auditable.

TRUST

Governed means proven — 42 / 42 checks pass

Data-quality is enforced across five dimensions and surfaced live in the app.

42
Total DQ checks

42
Checks passed

100%
Pass rate

DIMENSION	CHECKS	STATUS
Business Rules	11	100% PASS
Completeness	9	100% PASS
Uniqueness	9	100% PASS
Referential Integrity	7	100% PASS
Freshness	6	100% PASS

An alert writes to a pipeline-audit table if any check ever fails.

BUILT WITH COCO

CoCo drove every phase — end to end

From ontology design to a working app, with a full prompt log kept as evidence.

1. **Ontology & source design** — CoCo analyzed Marketplace datasets, defined entities, relationships, governed metrics, and the 24-table canonical source layer.
2. **Transformation layer** — CoCo generated the Bronze→Silver→Gold Dynamic Tables, UDFs, streams, tasks, and data-quality checks.
3. **Semantic view** — CoCo built **SV_SUPPLY_CHAIN** with governed definitions as metric descriptions and synonyms for entity resolution.
4. **Streamlit app** — CoCo generated the 3-tab Cortex Analyst app and iteratively fixed every runtime issue to a working deploy.

Evidence preserved: verbatim prompt logs across all four phases (29 documented steps), plus a screenshot checklist — demonstrating exactly how and where CoCo was used.

STATUS & ROADMAP

A complete, working backbone — with room to grow

✅ Built & verified

- 5-schema medallion, 1.03M rows, deterministic
- Governed semantic view (4 metrics, 18 dims)
- Cortex Analyst NL interface — verified answers
- 3-tab Streamlit app (Dashboard · Chat · Governance)
- 42/42 data-quality checks passing

🚀 Next

- Extend semantic view with **risk & ESG** (sanctions, spend-at-risk, ESG violations)
- Demand **forecasting** (Cortex ML) on the clean history series
- Business **alerts** — OTD threshold, stockout risk, supplier-risk escalation
- Cortex **Agent** blending metrics with unstructured documents

The backbone is done. Every next step plugs into the same governed foundation — no rework.

📦 Supply Chain Governed Analytics

One question. One trusted number. Every time.

A governed supply-chain ontology on Snowflake, a conversational interface powered by Cortex Analyst, and data quality you can prove — built end-to-end with CoCo.

4	100%	3	NL
governed metrics	DQ pass rate	app tabs	ask in English

Thank you — questions welcome.